SAFETY DATA SHEET

PROPYLENE GLYCOL USP/EP

Version	Revision Date:	SDS Number:	Date of last issue: 2019-10-22
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1. PRODUCT AND COMPANY IDENTIFICATION

Trade name	:	PROPYLENE GLYCOL USP/EP
CAS Number:	:	57-55-6
Chemical characterization	:	Glycols
Chemical name	:	1,2-Propanediol
Synonyms	:	Propylene Glycol, 1,2-Propanediol, 1,2-Dihydroxypropane, Monopropylene Glycol
Identified uses	:	Solvent; Intermediate; Functional Fluids
Prohibited uses	:	Active pharmaceutical ingredient (API); Tobacco; Electronic cigarettes (E-cigarettes); Cannabis; Theater fogs; Artificial smoke; Cat food; Sprinkler systems over 30%

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2. HAZARDS IDENTIFICATION

GHS Classification

Not a hazardous substance or mixture.

GHS label elements

Not a hazardous substance or mixture.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture	:	Substance
Substance name	:	1,2-Propanediol

CAS-No. : 200-338-0 (EINECS)

Components

Chemical name	CAS-No.	Concentration (% w/w)
Propylene Glycol	57-55-6	>= 99.5

4. FIRST AID MEASURES

General advice : May cause irritation of the eyes, skin and mucous membranes.
Always observe self-protection methods
Move out of dangerous area.
Remove contaminated shoes and clothing.
Show this material safety data sheet to the doctor in attendance.

If inhaled : Remove to fresh air.
In the case of inhalation of aerosol/mist consult a physician if necessary.
Not expected to present a significant inhalation hazard under anticipated conditions of normal use.
Avoid inhalation of hot vapors or extremely high concentrations of aerosols.

In case of skin contact : Wash skin thoroughly with mild soap and water.

In case of eye contact : Flush eyes with water thoroughly and continuously for 15 minutes.
Remove contact lenses, if present and easy to do. Continue rinsing.
If eye irritation persists, consult a specialist.

If swallowed : Not expected to present a significant ingestion hazard under anticipated conditions of normal use.

Most important symptoms and effects, both acute and delayed : High doses may cause CNS depression (fatigue, dizziness and possibly loss of concentration, with collapse, coma and death in cases of severe over-exposure).
May cause eye, skin, and respiratory tract irritation.

Notes to physician : Treat symptomatically.
Treatment of overexposure should be directed at the control of symptoms and the clinical condition of the patient.

5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : SMALL FIRE: Use dry chemicals, CO2, water spray or alcohol-resistant foam. LARGE FIRE: Use water spray, water fog

or alcohol-resistant foam.

Unsuitable extinguishing media	:	Do not use solid water stream.
Specific hazards during fire fighting	:	Heat from fire can generate flammable vapor. When mixed with air and exposed to ignition source, vapors can burn in open or explode if confined. Vapors may be heavier than air. May travel long distances along the ground before igniting and flashing back to vapor source. Fine sprays/mists may be combustible at temperatures below normal flash point. Fight fire from a safe distance/protected location. Heat may build enough pressure to rupture closed containers/spreading fire/increasing risk of burns/injuries. Use water spray/fog for cooling. Avoid frothing/steam explosion. Although water soluble, may not be practical to extinguish fire by water dilution. Notify authorities immediately if liquid enters sewer/public waters.
Special protective equipment for fire-fighters	:	Wear positive pressure self-contained breathing apparatus (SCBA). Structural firefighter's protective clothing will only provide limited protection.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Use personal protective equipment. Clean-up to be performed only by trained and properly equipped personnel.
Environmental precautions	:	Try to prevent the material from entering drains or water courses.
Methods and materials for containment and cleaning up	:	Extinguish ignition sources; stop release; prevent flow to sewers or public waters. Notify fire and environmental authorities. Impound/recover large land spill; soak up small spill with inert solids. Soak up small spills with inert solids. Use suitable disposal containers. On water, material is soluble and may float or sink. Contain/collect rapidly to minimize dispersion. Disperse residue to reduce aquatic harm. Report per regulatory requirements.

7. HANDLING AND STORAGE

Advice on protection against fire and explosion	:	Normal measures for preventive fire protection.
Advice on safe handling	:	Handle empty containers with care - residue can burn if heated. Empty containers should be thoroughly rinsed with copious amounts of clean water. The rinse water can be used for makeup water for any necessary dilution of the concentrated product before use, or it can be properly discarded.
Conditions for safe storage	:	Handle empty containers with care - residue may be combustible. Empty containers should be thoroughly rinsed with copious amounts of clean water. The rinse water can be used for makeup water for any necessary dilution of the concentrated product before use, or it can be properly discarded.
Materials to avoid	:	Carbon/Mild steel with suitable internal coating, or stainless steel
Further information on storage stability	:	No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Engineering measures	:	No special ventilation is recommended under anticipated conditions of normal use beyond that needed for normal comfort control.

Personal protective equipment

Respiratory protection	:	No special respiratory protection equipment is recommended under anticipated conditions of normal use.
Hand protection		
Material	:	Chemical resistant gloves made of butyl rubber or nitrile rubber category III according to EN 374.
Remarks	:	Not normally considered a skin hazard.
Eye protection	:	Use splash goggles when eye contact due to splashing or spraying liquid is possible.
Skin and body protection	:	No special clothing/skin protection equipment is recommended under normal conditions of anticipated use. Where use can result in skin contact, practice good personal

hygiene.

Hygiene measures : Selection of appropriate personal protective equipment should be based on an evaluation of the performance characteristics of the protective equipment relative to the task(s) to be performed, conditions present, duration of use, and the hazards and/or potential hazards that may be encountered during use.
Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure.
Use good personal hygiene practices.
Wash hands before eating, drinking, smoking, or using toilet facilities.
Take off contaminated clothing and wash before reuse.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	:	liquid (20 °C , 1,013.25 hPa)
Color	:	clear
Odor	:	Little or no odor.
Odor Threshold	:	No value available.
pH	:	no data available
Melting point/range	:	-60 °C
Boiling point/boiling range	:	184 °C (1003.20 hPa)
Flash point	:	104 °C (1000.010 hPa)
Flammability (solid, gas)	:	Not applicable
Self-ignition	:	> 400 °C
Upper explosion limit / Upper flammability limit	:	~ 17.4 vol%
Lower explosion limit / Lower flammability limit	:	~ 2.4 vol%
Vapor pressure	:	0.2 hPa (25 °C)

Relative vapor density	:	no data available
Density	:	1.03 g/cm3 (20 °C)
Solubility(ies) Water solubility	:	completely soluble (20 °C) pH: 7.1 - 7.8
Partition coefficient: n-octanol/water	:	log Pow: -1.07 (20.5 °C) pH: 6.2 - 6.4
Decomposition temperature	:	Incomplete combustion may produce carbon monoxide and other toxic gases.
Viscosity Viscosity, kinematic	:	42.1 mm2/s (25 °C)
Explosive properties	:	Not explosive
Oxidizing properties	:	The substance or mixture is not classified as oxidizing.
Surface tension	:	71.6 mN/m, 1.01 g/l, 21.5 °C

10. STABILITY AND REACTIVITY

Reactivity	:	Stable under recommended storage conditions.
Chemical stability	:	Stable under recommended storage conditions.
Possibility of hazardous reactions	:	Not expected to occur. This material is stable when properly handled and stored.
Conditions to avoid	:	High temperatures, oxidizing conditions. May degrade when exposed to light or other radiation sources.
Incompatible materials	:	Reacts with strong oxidizing agents. Strong acids. Isocyanates.
Hazardous decomposition products	:	Combustion may produce oxides of carbon and other toxic gases.

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Components:

Propylene Glycol:

Acute oral toxicity	:	LD50 Oral (Rat): 22,000 mg/kg
Acute inhalation toxicity	:	Assessment: The substance or mixture has no acute inhalation toxicity
Acute dermal toxicity	:	LD50 (Rabbit): > 5,000 mg/kg

Skin corrosion/irritation

Product:

Remarks	:	Based on skin irritation values, not classified. May cause slight transient skin irritation.

Components:

Propylene Glycol:

Assessment	:	Non-irritating to the skin.

Serious eye damage/eye irritation

Product:

Remarks	:	May produce minimal, fully reversible eye irritation.

Components:

Propylene Glycol:

Assessment	:	Non-irritating to the eyes.

Respiratory or skin sensitization

Product:

Test Type	:	Respiratory sensitization
Remarks	:	no data available
	:	Skin sensitization
Remarks	:	Skin reactions of unknown etiology have been described in some hypersensitive individuals following topical application.

Components:

Propylene Glycol:

Assessment : Not sensitizing

Germ cell mutagenicity

Product:

Germ cell mutagenicity - Assessment : Not classified

Components:

Propylene Glycol:

Germ cell mutagenicity - Assessment : Negative for genotoxicity using both in vitro and in vivo tests.

Carcinogenicity

Product:

Carcinogenicity - Assessment : Not classified

Components:

Propylene Glycol:

Carcinogenicity - Assessment : No increase in tumors was noted in rats and dogs exposed to high concentrations of propylene glycol via the diet for up to 2 years. The incidence of skin tumors was unaltered in mice following dermal application over a lifetime.

Reproductive toxicity

Product:

Reproductive toxicity - Assessment : Not classified

Components:

Propylene Glycol:

Reproductive toxicity - Assessment : No adverse effect on reproductive performance was seen in male and female mice exposed continuously to high doses of propylene glycol in drinking water for up to 3 months.
Results from studies in pregnant rats, mice, hamsters and rabbits demonstrate that propylene glycol is not teratogenic or fetotoxic.

STOT - single exposure

Product:

Assessment : Based on single exposure toxicity values, not classified.

Components:

Propylene Glycol:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

STOT - repeated exposure

Product:

Assessment : Based on repeated exposure toxicity values, not classified. Propylene glycol is of low inherent toxicity in rats and dogs after repeated oral exposure, while cats show species-specific hematological changes in red blood cells (other tissues unremarkable). Rats exposed repeatedly to high aerosol concentrations exhibited signs consistent with irritation of the eyes and nasal mucosa but showed no evidence of systemic toxicity.

Components:

Propylene Glycol:

Assessment : Long-term studies in rodents conducted with high oral doses found no evidence of adverse effects. Ingestion by cats, however, results in species-specific hematological changes.

Repeated dose toxicity

Components:

Propylene Glycol:

Species : Rat, male
NOAEL : 1,700 mg/kg
Application Route : Oral
Remarks : Long-term studies in rodents conducted with high oral doses found no evidence of adverse effects. Ingestion by cats, however, results in species-specific hematological changes.

Species : Rat
LOAEL : 0.16 mg/l
Application Route : Inhalation
Remarks : High aerosol concentrations inhaled by rats caused minor nasal and ocular signs that may have been due to mild irritation or drying effects on mucous membranes.

Further information

Product:

Remarks : no data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Ecotoxicology Assessment

Acute aquatic toxicity : Based on acute aquatic toxicity values, not classified.

Chronic aquatic toxicity : Not classified, based on readily biodegradability and low acute toxicity.

Components:

Propylene Glycol:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 40,613 mg/l
Exposure time: 96 HOURS

Toxicity to daphnia and other aquatic invertebrates : EC50 (Ceriodaphnia dubia): 18,340 mg/l
Exposure time: 48 HOURS

EC50 (Americamysis bahia (Mysid shrimp)): 18,800 mg/l
Exposure time: 96 HOURS

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (algae)): 19,000 mg/l
Exposure time: 96 HOURS

EC50 (Skeletonema costatum (marine diatom)): 19,100 mg/l
Exposure time: 96 HOURS

Toxicity to fish (Chronic toxicity) : Remarks: Not expected to exhibit chronic toxicity to fish.

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Ceriodaphnia dubia): 13,020 mg/l
Exposure time: 7 DAY

Toxicity to microorganisms : NOEC (Pseudomonas putida): 20,000 mg/l
Exposure time: 18 HOURS

Persistence and degradability

Components:

Propylene Glycol:

Biodegradability	:	Result: Biodegradable Biodegradation: 72 - 100 %

Bioaccumulative potential

Product:

Bioaccumulation	:	Remarks: This material is not expected to bioaccumulate.

Components:

Propylene Glycol:

Bioaccumulation	:	Bioconcentration factor (BCF): 0.09 Remarks: This material is not expected to bioaccumulate.
Partition coefficient: n-octanol/water	:	log Pow: -1.07 (20.5 °C)

Mobility in soil

Product:

Distribution among environmental compartments	:	Stability in soil Remarks: Low potential for soil adsorption expected

Components:

Propylene Glycol:

Distribution among environmental compartments	:	Remarks: Environmental releases of propylene glycol will tend to partition to water and soil, with little potential for evaporation.

Other adverse effects

Product:

Environmental fate and pathways	:	Remarks: No additional information available.
Results of PBT and vPvB assessment	:	Not applicable.

Components:

Propylene Glycol:

Environmental fate and pathways	:	This material is not expected to persist in the environment and should pose little if any physical or toxicological hazards.

Results of PBT and vPvB assessment	:	This substance is not considered to be persistent, bioaccumulating and toxic (PBT).
Additional ecological information	:	This material is expected to be non-hazardous to aquatic species.

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues	:	Comply with applicable local, state or international regulations concerning solid or hazardous waste disposal and/or container disposal. Landfill solids at permitted sites. Burn concentrated liquids, diluting with clean, low viscosity fuel. Dilute aqueous waste may biodegrade. Assure effluent complies with applicable regulations.

14. TRANSPORT INFORMATION

Not regulated for transport

BLG (MARPOL Annex II)

Description of the goods	:	PROPYLENE GYLCOL
Pollution category	:	OS
Ship type	:	NONE

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Global Inventory Status

The ingredients of this product are compliant with the following chemical inventory requirements or exemptions.

Country/Region	Inventory	Status Description
Australia	AICS	Listed
Canada	DSL	Listed
China	IECSC	Listed
Europe	REACH	See Compliance Statement*
Japan	ENCS	Listed
Korea	K REACH	Pre-registration period *
New Zealand	NZIoC	Listed
Philippines	PICCS	Listed

United Kingdom	UK REACH	See Compliance Statement*
United States of America	TSCA	Listed
Taiwan	TCSCA	Listed
Turkey	KKDIK	Pre-registration period *

* If the product has been purchased domestically from the notifying/registering legal entity of the LyondellBasell group of companies. We confirm that all substances (in this preparation) have been registered in accordance with the deadlines set forth in the applicable regulation. During the “Pre-registration period”, we confirm that all substances in this preparation have been pre-registered or, where required under the regulation, registered, and that we have the intention to proceed with their registration in accordance with the deadlines set forth in the regulation. For more information, please contact reach@lyondellbasell.com.

† For more information on the status of this material, please contact chemical control at global.chemical.control@lyondellbasell.com.

16. OTHER INFORMATION

Date format : dd.mm.yyyy

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evalua-

tion, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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